

BUS TICKETING APPLICATION - API DOCUMENTATION

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1. USER ROUTES

1.1 POST /users/register

Description: Registers a new user with the provided details.

Request Body:

```
{
  "name": "string",
  "phone": "string",
  "email": "string",
  "password": "string",
  "role": "string",
  "companyName": "string",
  "companyAddress": "string"
}
```

Response(201):

```
{
  "success": true,
  "message": "User registered successfully",
  "data": {}
}
```

1.2 POST /users/login

Description: Logs in a user with the provided credentials.

Request Body:

```
{
  "email": "string",
  "password": "string"
}
```

Response(200):

```
{
  "success": true,
  "message": "User Logged in successfully",
  "data": {"token": "string"}
}
```

1.3 POST /users/logout

Description: Logs out the current user.

Authentication: Required

Response(200):

```
{
  "success": true,
  "message": "User Logged out successfully"
}
```

1.4 GET /users/profile

Description: Retrieves the profile of the currently logged-in user.

Authentication: Required

Response(200):

```
{
  "success": true,
  "message": "Profile retrieved successfully",
  "data": {
    "id": "string",
    "name": "string",
    "email": "string",
    "phone": "string",
    "role": "string"
  }
}
```

1.5 PUT /users/profile

Description: Updates the profile of the currently logged-in user.

Authentication: Required

Request Body:

```
{
  "name": "string",
  "phone": "string"
}
```

Response(200):

```
{
  "success": true,
  "message": "Profile updated successfully",
  "data": {
    "id": "string",
    "name": "string",
    "email": "string",
    "phone": "string",
    "role": "string"
  }
}
```

2. BOOKING ROUTES

2.1 POST /bookings/user

Description: Creates a new booking for the current user.

Authentication: Required

Request Body:

```
{
  "tripId": "string",
  "seatNumbers": ["string"],
  "passengerDetails": [{
    "name": "string",
    "age": "number",
    "gender": "string"
  }]
}
```

Response(201):

```
{
  "success": true,
  "message": "Booking created successfully",
  "data": {
    "bookingId": "string",
    "status": "string",
    "totalAmount": "number"
  }
}
```

2.2 GET /bookings/user

Description: Gets all bookings for the current user.

Authentication: Required

Response(200):

```
{
  "success": true,
  "message": "User bookings retrieved
successfully",
  "data": [{
    "id": "string",
    "tripDetails": {},
    "seatNumbers": ["string"],
    "status": "string",
    "totalAmount": "number"
  }]
}
```

2.3 GET /bookings/operator

Description: Gets all bookings for the operator's buses.

Authentication: Required

Authorization: Operator, Admin

Response(200):

```
{
  "success": true,
  "message": "Operator bookings retrieved
successfully",
  "data": [{
    "id": "string",
    "userDetails": {},
    "tripDetails": {},
    "seatNumbers": ["string"],
    "status": "string"
  }]
}
```

2.4 GET /bookings

Description: Gets all bookings in the system.

Authentication: Required

Authorization: Admin

Response(200):

```
{
  "success": true,
  "message": "All bookings retrieved
successfully",
  "data": [{
    "id": "string",
    "userDetails": {},
    "tripDetails": {},
    "seatNumbers": ["string"],
    "status": "string"
  }]
}
```

2.5 GET /bookings/:id

Description: Gets a specific booking by ID.

Authentication: Required

Authorization: Admin

Response(200):

```
{
  "success": true,
  "message": "Booking retrieved successfully",
  "data": {
    "id": "string",
    "userDetails": {},
    "tripDetails": {},
    "seatNumbers": ["string"],
    "status": "string"
  }
}
```

2.6 PUT /bookings/:id

Description: Updates a specific booking.

Authentication: Required

Authorization: Admin

Request Body:

```
{
  "status": "string"
}
```

Response(200):

```
{
  "success": true,
  "message": "Booking updated successfully",
  "data": {
    "id": "string",
    "status": "string"
  }
}
```

2.7 DELETE /bookings/:id

Description: Deletes a specific booking.

Authentication: Required

Response(200):

```
{
  "success": true,
  "message": "Booking deleted successfully"
}
```

3. BUS ROUTES

3.1 POST /buses

Description: Creates a new bus.

Authentication: Required

Authorization: Operator, Admin

Request Body:

```
{
  "busNumber": "string",
  "busType": "string",
  "totalSeats": "number",
  "amenities": ["string"]
}
```

Response(201):

```
{
  "success": true,
  "message": "Bus created successfully",
  "data": {
    "id": "string",
    "busNumber": "string",
    "busType": "string",
    "totalSeats": "number"
  }
}
```

3.2 GET /buses

Description: Gets all buses.

Authentication: Required

Authorization: Operator, Admin

Response(200):

```
{
  "success": true,
  "message": "Buses retrieved successfully",
  "data": [{
    "id": "string",
    "busNumber": "string",
    "busType": "string",
    "totalSeats": "number"
  }]
}
```

3.3 GET /buses/operator

Description: Gets all buses owned by the operator.

Authentication: Required

Authorization: Operator

Response(200):

```
{
  "success": true,
  "message": "Operator buses retrieved
successfully",
  "data": [{
    "id": "string",
    "busNumber": "string",
    "busType": "string",
    "totalSeats": "number"
  }]
}
```

3.4 PUT /buses/:busId

Description: Updates a specific bus.

Authentication: Required

Authorization: Operator, Admin

Request Body:

```
{
  "busNumber": "string",
  "busType": "string",
  "totalSeats": "number",
  "amenities": ["string"]
}
```

Response(200):

```
{
  "success": true,
  "message": "Bus updated successfully",
  "data": {
    "id": "string",
    "busNumber": "string",
    "busType": "string",
    "totalSeats": "number"
  }
}
```

3.5 DELETE /buses/:busId

Description: Deletes a specific bus.

Authentication: Required

Authorization: Operator, Admin

Response(200):

```
{
  "success": true,
  "message": "Bus deleted successfully"
}
```

4. TRIP ROUTES

4.1 POST /trips

Description: Creates a new trip.

Authentication: Required

Authorization: Admin, Operator

Request Body:

```
{
  "busId": "string",
  "source": "string",
  "destination": "string",
  "departureTime": "datetime",
  "arrivalTime": "datetime",
  "fare": "number"
}
```

Response(201):

```
{
  "success": true,
  "message": "Trip created successfully",
  "data": {
    "id": "string",
    "busDetails": {},
    "source": "string",
    "destination": "string",
    "departureTime": "datetime",
    "arrivalTime": "datetime",
    "fare": "number"
  }
}
```

4.2 GET /trips

Description: Gets all trips.

Authentication: Required

Authorization: Admin, Operator

Response(200):

```
{
  "success": true,
  "message": "Trips retrieved successfully",
  "data": [{
    "id": "string",
    "busDetails": {},
    "source": "string",
    "destination": "string",
    "departureTime": "datetime",
    "arrivalTime": "datetime",
    "fare": "number"
  }]
}
```

4.3 GET /trips/search

Description: Gets trips based on filters.

Authentication: Required

Query Parameters:

```
source: "string"
destination: "string"
date: "date"
```

Response(200):

```
{
  "success": true,
  "message": "Filtered trips retrieved
successfully",
  "data": [{
    "id": "string",
    "busDetails": {},
    "source": "string",
    "destination": "string",
    "departureTime": "datetime",
    "arrivalTime": "datetime",
    "fare": "number",
    "availableSeats": ["string"]
  }]
}
```

4.4 GET /trips/:tripId

Description: Gets a specific trip by ID.

Authentication: Required

Response(200):

```
{
  "success": true,
  "message": "Trip retrieved successfully",
  "data": {
    "id": "string",
    "busDetails": {},
    "source": "string",
    "destination": "string",
    "departureTime": "datetime",
    "arrivalTime": "datetime",
    "fare": "number",
    "availableSeats": ["string"]
  }
}
```

4.5 PUT /trips/:tripId

Description: Updates a specific trip.

Authentication: Required

Authorization: Admin, Operator

Request Body:

```
{
  "source": "string",
  "destination": "string",
  "departureTime": "datetime",
  "arrivalTime": "datetime",
  "fare": "number"
}
```

Response(200):


```
{
  "success": true,
  "message": "Trip updated successfully",
  "data": {
    "id": "string",
    "source": "string",
    "destination": "string",
    "departureTime": "datetime",
    "arrivalTime": "datetime",
    "fare": "number"
  }
}
```

4.6 DELETE /trips/:tripId

Description: Deletes a specific trip.

Authentication: Required

Authorization: Admin, Operator

Response(200):

```
{
  "success": true,
  "message": "Trip deleted successfully"
}
```

5. RATING ROUTES

5.1 POST /ratings/bookings/:bookingId

Description: Submits a rating for a completed trip.

Authentication: Required

Authorization: User

Request Body:

```
{
  "rating": "number",
  "comment": "string"
}
```

Response(201):

```
{
  "success": true,
  "message": "Rating submitted successfully",
  "data": {
    "id": "string",
    "rating": "number",
    "comment": "string"
  }
}
```

5.2 PUT /ratings/:ratingId

Description: Updates an existing rating.

Authentication: Required

Authorization: User

Request Body:

```
{
  "rating": "number",
  "comment": "string"
}
```

Response(200):

```
{
  "success": true,
  "message": "Rating updated successfully",
  "data": {
    "id": "string",
    "rating": "number",
    "comment": "string"
  }
}
```

5.3 GET /ratings

Description: Gets all ratings submitted by the current user.

Authentication: Required

Authorization: User

Response(200):

```
{
  "success": true,
  "message": "User ratings retrieved successfully",
  "data": [{
    "id": "string",
    "bookingDetails": {},
    "rating": "number",
    "comment": "string"
  }]
}
```

5.4 DELETE /ratings/:ratingId

Description: Deletes a specific rating.

Authentication: Required

Authorization: User

Response(200):

```
{
  "success": true,
  "message": "Rating deleted successfully"
}
```

5.5 GET /ratings/buses/:busId

Description: Gets all ratings for a specific bus.

Authentication: Required

Authorization: Operator

Response(200):

```
{
  "success": true,
  "message": "Bus ratings retrieved
successfully",
  "data": [{
    "id": "string",
    "userDetails": {},
    "rating": "number",
    "comment": "string"
  }]
}
```

5.6 GET /ratings/bookings/:bookingId

Description: Gets the rating for a specific booking.

Authentication: Required

Authorization: Operator

Response(200):

```
{
  "success": true,
  "message": "Booking rating retrieved
successfully",
  "data": {
    "id": "string",
    "userDetails": {},
    "rating": "number",
    "comment": "string"
  }
}
```

6. ADMIN ROUTES

6.1 PUT /admin/block/:userId

Description: Blocks a user from using the application.

Authentication: Required

Authorization: Admin

Response(200):

```
{
  "success": true,
  "message": "User blocked successfully",
  "data": {
    "id": "string",
    "name": "string",
    "email": "string",
    "isBlocked": true
  }
}
```

6.2 PUT /admin/unblock/:userId

Description: Unblocks a previously blocked user.

Authentication: Required

Authorization: Admin

Response(200):

```
{
  "success": true,
  "message": "User unblocked successfully",
  "data": {
    "id": "string",
    "name": "string",
    "email": "string",
    "isBlocked": false
  }
}
```

I'll create API documentation for the admin routes we've implemented, following the format you provided. Here's the API documentation for all the admin routes:

6. Admin Routes

6.1 GET /admin/users

Description: Retrieves a list of all users, with optional filters.

Authentication: Required

Authorization: Admin

Query Parameters:

- `role` (optional): Filter users by role (user, operator, admin)
- `isBlocked` (optional): Filter by blocked status (true, false)

Response(200):

```
{
  "success": true,
  "message": "Users retrieved successfully",
  "data": [
    {
      "id": "string",
      "name": "string",
      "email": "string",
      "phone": "string",
      "role": "string",
      "isBlocked": boolean,
      "isVerified": boolean
    }
  ]
}
```

6.2 PUT /admin/block/:userId

Description: Blocks a user, preventing them from logging in.

Authentication: Required

Authorization: Admin

Response(200):

```
{
  "success": true,
  "message": "User blocked successfully",
  "data": {
    "id": "string",
    "name": "string",
    "email": "string",
    "phone": "string",
    "isBlocked": true
  }
}
```

6.3 PUT /admin/unblock/:userId

Description: Unblocks a previously blocked user.

Authentication: Required

Authorization: Admin

Response(200):

```
{
  "success": true,
  "message": "User unblocked successfully",
  "data": {
    "id": "string",
    "name": "string",
    "email": "string",
    "phone": "string",
    "isBlocked": false
  }
}
```

6.4 PUT /admin/verify-operator/:userId

Description: Verifies an operator account, allowing them to add buses and create trips.

Authentication: Required

Authorization: Admin

Request Body:

```
{
  "verificationStatus": boolean
}
```

Response(200):

```
{
  "success": true,
  "message": "Operator verification status updated successfully",
  "data": {
    "id": "string",
    "name": "string",
    "email": "string",
    "role": "operator",
    "isVerified": boolean
  }
}
```

6.5 GET /admin/trips

Description: Retrieves a list of all trips, with optional filters.

Authentication: Required

Authorization: Admin

Query Parameters:

- status (optional): Filter trips by status (active, completed, cancelled)
- from (optional): Filter by departure location
- to (optional): Filter by arrival location
- date (optional): Filter by departure date (YYYY-MM-DD)

Response(200):

```
{
  "success": true,
  "message": "Trips retrieved successfully",
  "data": [
    {
      "id": "string",
      "from": "string",
      "to": "string",
      "departureTime": "datetime",
      "arrivalTime": "datetime",
      "fare": number,
      "availableSeats": number,
      "status": "string",
      "bus": {
        "id": "string",
        "regNumber": "string",
        "type": "string",
        "operator": "string"
      }
    }
  ]
}
```

6.6 GET /admin/bookings

Description: Retrieves a list of all bookings, with optional filters.

Authentication: Required

Authorization: Admin

Query Parameters:

- `status` (optional): Filter bookings by status (confirmed, cancelled, completed)
- `paymentStatus` (optional): Filter by payment status (paid, pending, refunded)
- `tripId` (optional): Filter bookings by specific trip
- `userId` (optional): Filter bookings by specific user

Response(200):

```
{
  "success": true,
  "message": "Bookings retrieved successfully",
  "data": [
    {
      "id": "string",
      "trip": {
        "id": "string",
        "from": "string",
        "to": "string",
        "departureTime": "datetime",
        "arrivalTime": "datetime",
        "fare": number,
        "bus": {
          "regNumber": "string",
          "type": "string",
          "operator": "string"
        }
      },
      "user": {
        "id": "string",
        "name": "string",
        "email": "string",
        "phone": "string"
      },
      "seatNumbers": ["string"],
      "status": "string",
      "paymentStatus": "string",
      "bookingDate": "datetime",
      "totalAmount": number
    }
  ]
}
```

6.7 PUT /admin/trips/:tripId

Description: Modifies trip details in emergency situations.

Authentication: Required

Authorization: Admin

Request Body:

```
{
  "departureTime": "datetime",
  "arrivalTime": "datetime",
  "fare": number,
  "status": "string",
  "availableSeats": number
}
```

Response(200):

```
{
  "success": true,
  "message": "Trip modified successfully",
  "data": {
    "id": "string",
    "from": "string",
    "to": "string",
    "departureTime": "datetime",
    "arrivalTime": "datetime",
    "fare": number,
    "availableSeats": number,
    "status": "string"
  }
}
```

6.8 PUT /admin/trips/:tripId/cancel

Description: Cancels a trip in emergency situations and automatically cancels all associated bookings.

Authentication: Required

Authorization: Admin

Request Body:

```
{
  "reason": "string"
}
```

Response(200):

```
{
  "success": true,
  "message": "Trip cancelled successfully",
  "data": {
    "id": "string",
    "from": "string",
    "to": "string",
    "departureTime": "datetime",
    "arrivalTime": "datetime",
    "status": "cancelled",
    "cancellationReason": "string",
    "adminCancelled": true
  }
}
```